

SE – Lab 3

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Architecture Selection – Virtual Technical Conference Platform

Architectural Choice

We chose **Microservices Architecture** for the Virtual Technical Conference Platform.

Reason 1 – Separation of Functional Areas

The platform contains distinct functions such as manuscript management, reviewer assignment, conference scheduling, and live Q&A. Dividing these functions into separate services makes each part easier to develop, maintain, and modify independently.

Reason 2 – Independent Handling of Workloads

The platform has different performance requirements. Conference schedule generation must be completed within **10 seconds**, while the live Q&A queue must be updated within **2 seconds for at least 95% of updates** under the defined load. Separate services allow these different workloads to be handled independently.

Security Advantage

An **Authentication Service** can verify users and their roles before allowing access to system functions. This helps ensure that operations are restricted according to roles such as Conference Author, Track Chair, Reviewer, and Session Participant.

Performance Benefit

The **Live Q&A Service** can be optimized for frequent question and upvote requests, while the **Scheduling Service** can independently handle schedule-generation operations. This prevents one workload from unnecessarily affecting another.

Conclusion

Although Microservices Architecture introduces additional communication complexity, its separation of services, independent handling of workloads, and fault isolation make it suitable for the Virtual Technical Conference Platform.